

LOUGH FEA, United Kingdom

WMO: 039110

Lat: 54.7211N

Lon: 6.8147W

Elev: 745

StdP: 14.30

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.4	29.0	22.8	17.6	29.2	25.3	19.9	32.0	37.5	40.5	31.8	40.8	3.3	330	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	70.6	61.9	67.4	60.3	64.7	58.9	63.6	68.2	61.9	65.5	60.2	63.3	8.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.8	84.8	65.2	60.3	80.3	63.4	58.9	76.2	61.6	29.2	68.1	28.0	65.6	26.9	63.4	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.7	22.0	19.0	DB	21.7	74.9	4.1	3.3	18.8	77.3	16.4	79.3	14.1	81.1	11.2	83.5
			WB	20.6	66.5	4.4	2.1	17.5	68.0	14.9	69.2	12.4	70.3	9.2	71.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	38.5	38.9	40.9	44.5	49.4	54.1	57.0	56.6	53.3	48.2	42.7	39.3	(6)
	DBStd	8.12	4.83	4.59	4.64	4.50	4.87	4.38	4.04	3.61	3.77	4.38	4.63	5.40	(7)
	HDD50	1853	356	310	282	175	69	10	0	0	12	85	221	333	(8)
	HDD65	6571	821	730	746	616	482	327	250	261	352	520	668	798	(9)
	CDD50	763	0	0	1	9	52	133	218	207	110	30	3	0	(10)
	CDD65	5	0	0	0	0	0	0	3	2	0	0	0	0	(11)
	CDH74	25	0	0	0	0	0	7	13	5	0	0	0	0	(12)
	CDH80	1	0	0	0	0	0	1	0	0	0	0	0	0	(13)
(14) Wind	WSAvg	8.9	10.1	9.9	9.5	8.7	8.5	8.1	8.0	8.0	8.3	8.9	9.4	9.6	(14)
(15) Precipitation	PrecAvg	45.4	4.6	3.5	3.2	3.0	3.3	3.2	3.2	3.8	3.4	4.6	4.8	4.8	(15)
	PrecMax	55.9	7.2	6.9	5.7	4.8	5.6	6.5	4.6	7.9	7.2	8.4	9.0	9.6	(16)
	PrecMin	35.1	1.2	1.2	1.2	1.5	0.8	1.4	1.7	0.9	0.6	1.5	2.0	2.3	(17)
	PrecStd	5.2	1.6	1.5	1.1	1.0	1.3	1.3	1.1	1.6	1.8	2.0	1.8	1.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.6	52.5	56.4	62.2	70.2	73.0	76.0	73.8	67.2	60.1	55.7	53.4	(19)
		MCWB	50.5	50.4	49.6	53.9	60.0	61.4	65.7	65.2	61.7	57.5	54.6	52.7	(20)
	2%	DB	49.5	50.1	52.1	57.9	65.5	68.5	71.0	68.3	63.9	57.6	53.3	51.1	(21)
		MCWB	48.7	48.8	48.5	50.8	56.9	60.0	62.6	61.7	59.5	55.9	52.4	50.2	(22)
	5%	DB	47.8	48.2	49.9	54.5	61.4	65.2	67.3	65.1	61.5	56.2	51.4	49.2	(23)
		MCWB	46.9	47.1	47.4	49.4	54.3	58.0	60.8	60.0	58.0	54.6	50.6	48.5	(24)
	10%	DB	46.1	46.4	48.2	52.0	57.9	62.0	64.1	62.7	59.4	54.8	49.9	47.2	(25)
		MCWB	45.2	45.2	46.0	48.0	52.4	56.5	59.1	58.8	56.5	53.3	49.0	46.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.8	51.2	51.6	55.1	61.2	63.8	66.6	65.6	63.4	58.6	54.9	52.8	(27)
		MCDB	51.2	51.9	54.0	59.1	68.8	70.5	73.5	72.1	65.5	59.3	55.4	53.1	(28)
	2%	WB	48.9	49.2	49.6	52.5	57.8	61.1	63.9	63.1	60.8	56.6	52.7	50.4	(29)
		MCDB	49.4	49.8	51.1	55.9	63.7	66.5	69.3	66.4	62.6	57.3	53.2	50.9	(30)
	5%	WB	47.3	47.3	48.1	50.3	55.5	59.1	61.9	61.3	58.8	55.1	50.9	48.7	(31)
		MCDB	47.8	48.1	49.5	53.4	60.0	63.5	65.8	63.9	60.6	55.9	51.4	49.2	(32)
	10%	WB	45.4	45.4	46.3	48.5	53.3	57.4	60.1	59.7	57.2	53.6	49.2	46.6	(33)
		MCDB	46.1	46.3	47.9	51.3	57.2	61.3	63.3	61.9	58.9	54.5	49.7	47.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.9	10.4	11.9	13.1	12.5	11.8	11.3	10.9	9.3	8.2	7.9	(35)
		MCDBR	9.3	9.6	12.5	15.9	18.3	17.8	16.8	14.8	13.5	10.0	8.9	9.6	(36)
	5% WB	MCWBR	9.3	9.0	9.3	10.2	11.1	10.5	9.8	8.9	9.3	8.3	8.8	9.7	(37)
		MCWBR	9.3	9.2	11.0	14.1	16.3	15.5	14.9	12.8	11.8	9.1	8.9	9.7	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.333	0.343	0.366	0.378	0.383	0.373	0.375	0.359	0.353	0.340	0.326	0.321	(40)
		taud	2.204	2.286	2.263	2.270	2.297	2.359	2.357	2.417	2.430	2.395	2.267	2.171	(41)
	Ebn at Noon	Ebn at Noon	183	231	253	268	273	277	274	272	258	231	186	157	(42)
		Edh at Noon	20	26	33	37	38	36	36	32	28	23	18	16	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	181	397	725	1175	1491	1519	1392	1133	816	466	224	134	(44)
	RadStd	RadStd	19	35	70	112	135	153	115	94	42	42	24	13	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page